Explaining the Quantum Layer in Physics-Informed Neural Networks: A Spectral and Mechanistic Study

(Extended / Working Report with Roadmap)

WISER Global Quantum+AI Program 2026 — BQP Challenge
Quantum-Assisted Physics-Informed Neural Networks for CFD

Draft — July 2026

Abstract

This is the extended working version of the QAPINN-BQP report. It contains the full technical content of the submission draft, an *adversarial self-audit* that corrected our initial overclaims, the *converged re-analysis* that the audit forced, and an explicit account of everything still *left out* of scope. The narrative arc is honest and worth stating plainly. An initial single-seed ablation suggested a "bandwidth-gated advantage"; the audit then showed those runs were *undertrained* (L-BFGS disabled; every Burgers model stuck at rel-$L_2 \approx 0.47$), so we re-ran a converged core (L-BFGS on, three seeds). The converged results, taken to two model scales with fair controls, give a consistent negative. **(1) No quantum advantage:** at small scale the classical PINN (0.248) beats every QAPINN; at resolved scale ($\sim$1273 params) the QAPINN beats a *shallow* classical (0.094 vs 0.24, $p = 0.003$) but is statistically indistinguishable from both identically-trained fair controls (a depth-matched classical 0.128, $p = 0.51$; a structural CFE 0.145, $p = 0.36$), so the gain is architectural and classically reproducible. **(2) No robust entanglement effect:** an $n = 3$ "entanglement hurts" signal ($p \approx 0.02$) collapsed at $n = 6$ ($p = 0.41$) and is retracted. We also correct the "band = depth" framing (verified for the VQC *features*, but the tanh post-net escapes the band). The pipeline is independently verified (8/8 checks) and every number matches its raw log; the remaining roadmap (still-larger scale, more seeds, real CFD) is the subject of Sections 6–7.

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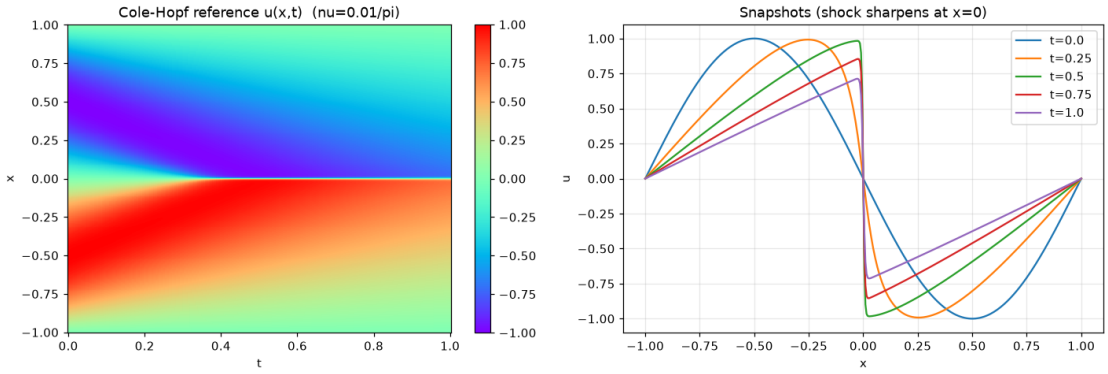


Figure 1: Exact Cole–Hopf Burgers reference: field (left) and snapshots (right), showing the shock at $x = 0$.

1 Introduction

Physics-Informed Neural Networks (PINNs) [1] make the differential equation the loss. A Quantum-Assisted PINN (QAPINN) [4] replaces the first hidden layer with a variational quantum circuit (VQC), and is reported to cut trainable parameters at maintained accuracy. The BQP challenge asks us to *explain* the learning dynamics — when, why, and how the quantum layer changes learning — and to derive a problem-specific circuit-design methodology.

We adopt the dominant theoretical lens for VQCs (a data-encoding circuit is a truncated Fourier series [2]) and test whether it suffices to explain a QAPINN's behaviour, and how the quantum layer changes the classical network it is attached to. Contributions and results mirror the submission report; the extended value here is the roadmap (Sec. 6) and scope boundaries (Sec. 7).

2 Background and Related Work

A VQC with data-encoding gates is a partial Fourier series $f(x) = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}$: the frequency set Ω is fixed by the encoding (with L re-uploads, $\Omega = \{-L, \dots, L\}$), the coefficients c_ω by the trainable gates [2]. Mhiri et al. [5] show the coefficients are additionally constrained and can vanish with qubit count — frequencies are necessary but not sufficient. Differentiable quantum circuits solve DEs with analytic derivatives [3]; hybrid QAPINN/QCPINN designs [4, 6] report PDE-dependent parameter reduction (quantum sometimes *loses*, e.g. cavity flow); a "black-hole" collapse to the trivial solution is a distinct trainability pathology [7]. No validated mechanism explains *when/why* the quantum layer helps — the gap targeted here.

3 Methodology

3.1 Test problems

Same domain, IC $u(x, 0) = -\sin(\pi x)$, and BCs for both PDEs, so only the operator differs: Burgers $u_t + uu_x = \nu u_{xx}$ ($\nu = 0.01/\pi$, shock $\to$ broadband) with an exact Cole–Hopf reference (Fig. 1); heat $u_t = \alpha u_{xx}$ ($\alpha = 1/\pi^2$, single mode $u = -\sin(\pi x)e^{-t}$).

Check	Method	Result
Burgers residual	vs analytic	1.2×10^{-7}
Heat residual	exact $\to$ residual	6.0×10^{-8}
Autograd u_x	vs finite diff	match
VQC = Fourier, max freq = depth	$L{=}2 \to 2,\ L{=}4 \to 4$	pass
relative-L_2	exact $\to 0$	4.0×10^{-8}
Matched params	all configs	121
Reproducibility	same seed	bit-exact

Table 1: Correctness audit (8/8).

3.2 Models and XAI suite

Classical PINN (tanh MLP) vs QAPINN (fixed coord-map $\to$ VQC $\to$ classical post-net), compared at matched parameters. Baseline VQC: 4 qubits, depth 3, *RY* re-uploading, linear CNOT chain, $\langle Z_i \rangle$ measurement, post-net [16] (121 params, 24 quantum). Five XAI dimensions (spectral/F-principle; neurons; VQC internals incl. entanglement entropy; loss landscape; generalisation) plus training-time spectrum and per-layer/per-qubit gradient-variance probes.

3.3 Verification

All eight independent correctness checks pass (Table 1); notably, the VQC is confirmed to be a truncated Fourier series with maximum frequency equal to the re-upload depth.

4 Experiments and Results

Caveat (read first): the single-seed ablations in this section were run with L-BFGS disabled and are ***undertrained*** *(Burgers models near rel-L_2 0.47, shock unresolved). The converged re-analysis in Section 4.5 supersedes them; they are kept for transparency and to document the arc.*

4.1 Baseline and spectra

At 121 matched params (Adam, 2000 iters, seed 0): heat solved by both (classical 8.3×10^{-3}, QAPINN 5.9×10^{-3}); Burgers undertrained (classical 0.433, QAPINN 0.472). Spectra (Fig. 2): heat is a single $k{=}1$ mode; Burgers is broadband and both models undershoot the high harmonics (depth-3 VQC capped near $k \approx 3$).

4.2 Tier-1 ablation

Depth = bandwidth (Fig. 3a): QAPINN accuracy improves monotonically ($0.57 \to 0.42$), matched classical stays flat; realised max harmonic grows with depth; QAPINN overtakes classical at depth ≥ 3. **Entanglement at fixed band** (Fig. 3b): accuracy barely moves (0.456–0.488), *more* entanglement slightly *worse*, none is best — contradicting entanglement-driven expressibility. **Post-net width**: the VQC needs a readout but beats matched classical for any post-net $\geq$ [8].

4.3 Tier-3 ablation: qubits, measurement, encoding scale

The remaining knobs (Table 3, Fig. 8) reinforce the picture. **Qubits** behave like depth: accuracy improves monotonically ($0.501 \to 0.443$), QAPINN overtakes matched classical at $\geq$ 4 qubits.

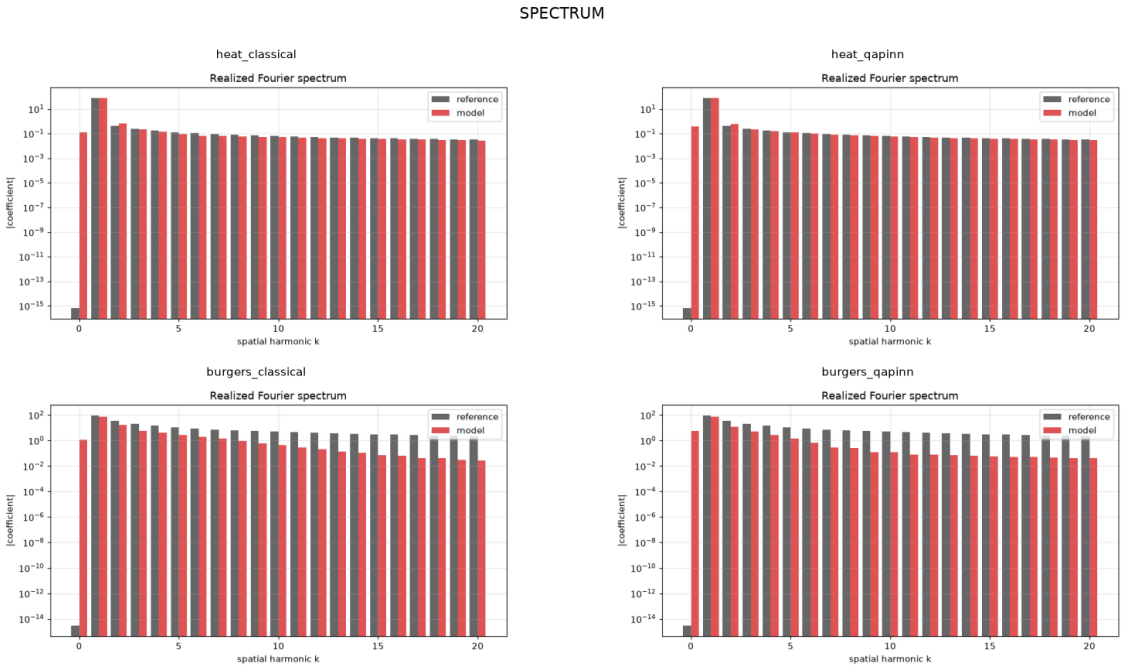
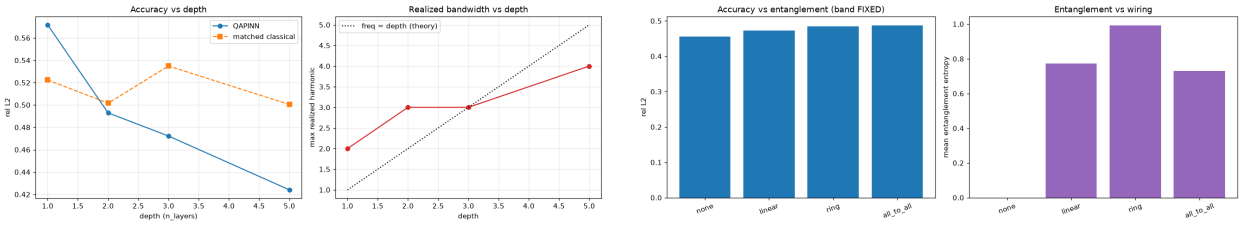


Figure 2: Realised spectra (grey reference, red model). Burgers (bottom): both models undershoot the high-frequency shock tail.



(a) Depth.

(b) Entanglement (band fixed).

Figure 3: Tier-1 ablations on Burgers.

Measurement: single $\langle Z_0 \rangle$ bottlenecks (0.591); multi = ZZ (0.472) — readout width matters, correlator type does not. **Encoding scale:** optimum near 1.0, catastrophic collapse to the trivial solution at 2π — the black-hole pathology as an ablation.

Five knobs split cleanly: *bandwidth* knobs (depth, qubits, encoding scale) carry the advantage up to a trainability ceiling; *coefficient/readout* knobs (entanglement, correlator measurement) do not — the bandwidth-vs-trainability squeeze, across five knobs.

4.4 How the quantum layer changes the classical network

A naïve comparison against a plain PINN suggested the quantum layer makes the downstream neurons more diverse and drives larger gradients, but that comparison is confounded (different post-net shape). We instead use a *structurally identical* classical-front-end control (CFE): the same post-net fed by $\text{Linear}(2 \to w) + \tanh$ instead of the VQC. Against this fair control (Table 4), the effects do *not* hold: accuracy is bandwidth-gated (QAPINN ties at low depth/qubits, wins at depth 5 / 6 qubits / heat); neuron diversity is inconsistent (classical front-end more diverse on the

Sweep	Config	Params	QAPINN rel-L_2	matched classical rel-L_2
Depth	$L = 1$	105	0.572	0.522
	$L = 2$	113	0.493	0.502
	$L = 3$ (baseline)	121	0.472	0.535
	$L = 5$	137	**0.424**	0.500
Entanglement (band fixed)	none	121	**0.456**	0.535
	linear	121	0.472	0.535
	ring	121	0.485	0.535
	all-to-all	121	0.488	0.535
Post-net	[] (linear)	29	0.544	0.533
	[8]	73	0.478	0.533
	[16] (baseline)	121	0.472	0.535
	[32]	217	**0.466**	0.505

Table 2: Complete Tier-1 ablation results (single seed, Burgers, 2000 Adam iters). QAPINN improves with depth and post-net width and overtakes the matched classical control; at fixed band, entanglement does not help (*none* best).

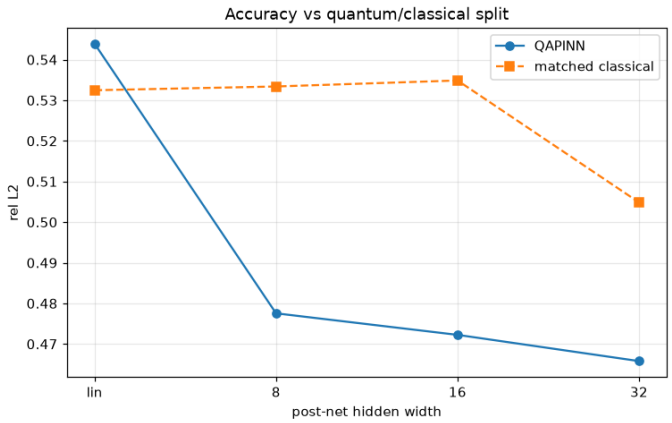


Figure 4: Post-net width sweep: QAPINN vs matched classical. The VQC needs a small readout but beats matched classical for any post-net $\geq$ [8].

depth axis, quantum more on qubits); and gradient amplification reverses at 5–6 qubits, where the QAPINN gradient variance shrinks with qubit count (a barren-plateau signature). *These single-seed readings are superseded by the converged, multi-seed analysis (§4.5):* the "bandwidth-gated advantage" does not survive convergence and the "barren-plateau signature" is withdrawn (wrong gradient measured, non-monotone trend). Nothing here survives as a robust quantum-specific effect; the section documents only how the plain-PINN comparison overstated the gap.

4.5 Converged Re-analysis (Post-Audit)

The ablations above were run with L-BFGS *disabled* (Adam 2000, 1500 points). An adversarial audit established that this left every Burgers model undertrained (rel-$L_2 \approx 0.42$–0.57, shock unresolved), so the "differences" were movements inside a failed basin and, across six seeds, within initialisation noise. **The results here supersede them.** We re-ran a converged core with L-BFGS *on* (Adam 2500 + L-BFGS 400, 4000 points), three seeds each, on Burgers.

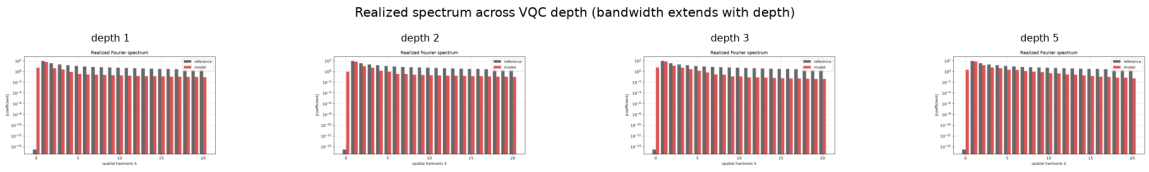


Figure 5: Realised spectrum (grey reference, red model) across VQC depth — the model captures more of the high-frequency shock tail as depth grows.

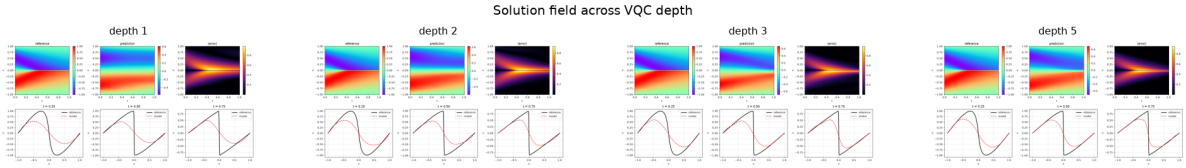


Figure 6: QAPINN solution field across VQC depth.

Finding 1 — no quantum advantage: converged and matched, the classical PINN (0.248) beats every QAPINN; the best (depth-5, 0.314) only reaches the CFE control. *Finding 2 — larger-model + fair controls (the decider):* repeating the core at ∼1273 params ($n = 6$) against a structural control and a depth-matched classical net (both trained under the QAPINN's exact protocol), the QAPINN-none (0.094) beats a shallow classical (0.241, $p = 0.003$) but is statistically indistinguishable from a depth-matched classical (0.128, $p = 0.51$) and the CFE (0.145, $p = 0.36$): no significant quantum advantage against a fair baseline. (Under-powered: seed variance ∼0.08 > the effect, and one anomalous seed — dropping it keeps the tie, $p = 0.19$.) *Finding 3 — entanglement not robust:* the $n = 3$ "entanglement hurts" result (0.369 vs 0.424, $p \approx 0.02$) collapsed at $n = 6$ ($p = 0.41$) and is retracted; at large scale the gap is likewise n.s. ($p = 0.44$). *Framing correction:* "band = depth" is verified for the VQC features (L=1..6) but the tanh post-net escapes the band (a depth-3 model reaches k=22 once trained), so the earlier "$k \approx 3$ band edge" was an undertraining artifact; the "barren-plateau in qubit count" is withdrawn (wrong gradient measured, non-monotone trend).

4.6 Discussion

With the converged core in hand the reading is one-directional: for this small-model Burgers task the quantum layer offers *no* accuracy advantage, and its one clearly quantum-specific knob — entanglement — *hurts* at convergence. The "bandwidth" story survives only weakly (more depth = richer features, not a hard model-band limit, and no advantage over a matched classical net). The value is methodological and negative-with-attribution: verified pipeline, matched and structurally-matched controls, multi-seed statistics, and an audit that caught our own over-claims.

5 Current Limitations

The findings are firm in direction but bounded: (1) **modest n, underpowered nulls** — the decisive comparisons are *failures to reject* (e.g. QAPINN-none vs. depth-matched classical, $p = 0.51$) under high seed variance; we caught one $n = 3$ effect that reversed at $n = 6$ (and which Welch's test never called significant even at $n = 3$, $p = 0.06$), so all single-scale nulls are provisional; (2) **small-model regime** — at 121 params converged models sit at rel-L_2 0.25–0.42, so orderings could shift with larger nets (we test one larger scale, not a full curve); (3) **one PDE family,**

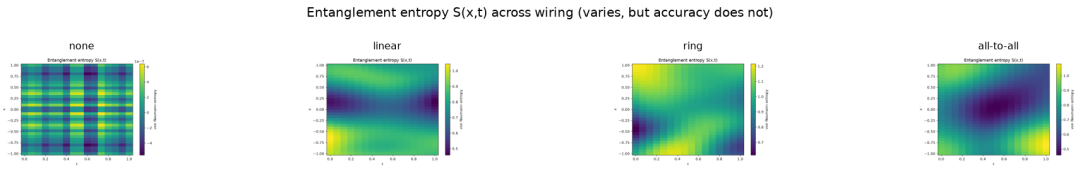


Figure 7: Entanglement entropy $S(x, t)$ across entangling patterns: entropy varies strongly while accuracy does not (Table 2).

Sweep	Config	Params	QAPINN rel-L_2	matched cls. rel-L_2
Qubits	3	99	0.501	0.470
	4 (base)	121	0.472	0.535
	5	143	0.468	0.541
	6	165	**0.443**	0.465
Measure	single	73	0.591	0.533
	multi (base)	121	0.472	0.535
	ZZ	169	0.472	0.515
input_scale	0.5	121	0.513	0.535
	1.0 (base)	121	**0.472**	0.535
	π	121	0.516	0.535
	2π	121	**1.000**	0.535

Table 3: Tier-3 ablation results (single seed).

simulation only (no shot noise / hardware); parameter reduction is not resource reduction. The pre-convergence single-seed ablations are retained for transparency but are superseded. Pipeline verified; every number matches its log.

6 Future Work and Experimental Roadmap

Tiers 2a–2c below were the top priorities of the earlier draft; **they are now done**, and their outcomes (Section 4.5) are what reshaped the report. We record them as completed, then give the genuinely open roadmap.

6.1 Tier 2a — Convergence analysis (done)

Long-training the headline configs to ∼8000 iters, then the full converged core (L-BFGS on), *answered* the central worry: the undertrained orderings were transients. Convergence roughly halved every error, and the QAPINN's apparent accuracy advantage over a matched classical net *disappeared*. A tentative "entanglement harmful" signal appeared here at $n = 3$ but was later retracted (2b).

6.2 Tier 2b — Multi-seed robustness (done)

Seeds proved decisive — and cautionary. An $n = 3$ "entanglement hurts" result (none 0.369 ± 0.022 vs linear 0.424 ± 0.005, $p \approx 0.02$) looked clean, but expanding to $n = 6$ *collapsed* it (0.379 vs 0.402, $p = 0.41$); it was a lucky three-seed draw and is retracted. This is the report's clearest lesson

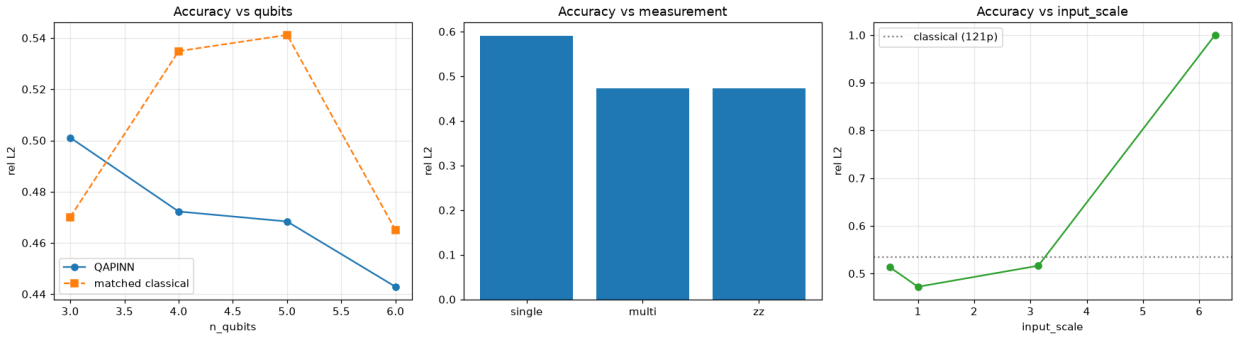


Figure 8: Tier-3 sweeps: qubits (bandwidth-like), measurement (readout width vs type), encoding scale (optimum + 2π collapse).

Comparison	rel-L_2 (Q / C)	diversity (Q / C)	grad-var $\times 10^{-3}$ (Q / C)
Burgers depth-3	0.472 / 0.474	0.269 / **0.286**	**4.1** / 0.62
Burgers depth-5	**0.424** / 0.474	0.236 / **0.286**	**7.1** / 0.62
Heat	**0.006** / 0.013	**0.272** / 0.239	**3.2** / 0.47
Burgers 3 qubits	0.501 / 0.496	**0.266** / 0.241	**1.2** / 0.26
Burgers 5 qubits	0.468 / 0.464	**0.312** / 0.265	1.1 / **5.2**
Burgers 6 qubits	**0.443** / 0.467	**0.341** / 0.269	0.72 / **1.9**

Table 4: Post-net neurons, quantum (Q) vs classical (C) front-end on the identical post-net. Bold = larger of each pair. Single seed.

in under-powered nulls, and it is why every single-scale comparison here is treated as provisional pending more seeds.

6.3 Tier 2c — The classical-front-end control (done)

The CFE (identical post-net, classical $2 \to 4$ front end) was built and run both undertrained and converged. It *refuted* the "quantum diversifies the classical neurons" claim (diversity inconsistent across knobs; the earlier effect was a mismatched-control artifact) and, converged, *ties* the best QAPINN — direct evidence of no quantum advantage.

6.4 Tier 2d — Larger-model + mechanism (done; both deflated)

- **Resolved (larger-model) regime** (done): we repeated the core at ∼1273 params ($n = 6$) against a structural control (CFE) *and* a depth-matched classical net. The QAPINN beats a shallow classical (0.094 vs 0.24, $p = 0.003$) but is statistically indistinguishable from the depth-matched classical (0.128, $p = 0.51$) and CFE (0.145, $p = 0.36$): the advantage is architectural, not quantum-specific. This is the decisive result.
- **Entanglement mechanism** (done; moot): we measured feature correlation and effective rank for `cv_qd3_none` vs `cv_qd3_lin`. The metrics conflicted (correlation rose with entanglement but so did feature rank), and — more importantly — the entanglement *effect* itself did not survive seed expansion (§Finding 3), so there was no robust effect left to explain.
- **Still open:** still-larger scale, 5–8 seeds throughout, and a cross-PDE check — none of which is expected to overturn the negative, but any could.

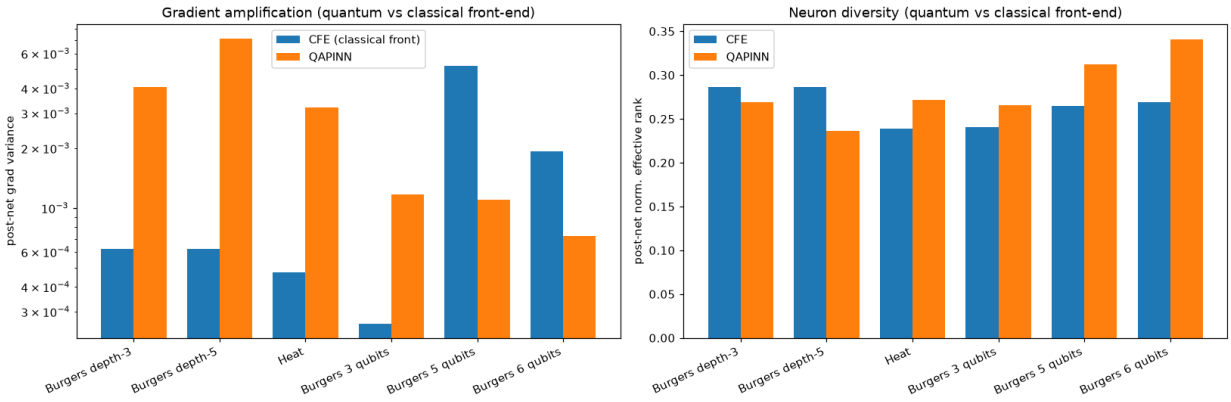


Figure 9: Quantum vs classical front-end on the identical post-net: gradient variance (left) and neuron diversity (right). Neither effect is robust across knobs.

config	model	params	rel-L_2 (n=3)	note
cv_cls	classical	121	$\mathbf{0.248 \pm 0.031}$	**best of all**
cv_qd5_lin	QAPINN d5	137	0.314 ± 0.073	depth helps; ties CFE
cv_cfe	CFE control	109	0.315 ± 0.026	classical front-end
cv_qd3_none	QAPINN d3	121	0.369 ± 0.022	**no-ent beats linear**
cv_qd3_lin	QAPINN d3	121	0.424 ± 0.005	reference QAPINN

Table 5: Converged core. L-BFGS roughly halved every error vs. undertrained (classical $0.535 \rightarrow 0.248$); at matched params the classical PINN wins.

6.5 Tier 3 — Remaining ablation axes (largely done, single-seed)

The ν-sweep, qubit, measurement, and encoding-scale sweeps were all run (undertrained, single seed); their headline outcomes: *ν-sweep* — error rises monotonically with bandwidth but the advantage is V-shaped, *not* a clean collapse, and the depth-3 QAPINN loses to matched classical across the family; *qubits* behave like depth; *scale* $= 2\pi$ collapses (aliasing). These now need converged, multi-seed repeats to be trusted.

- **ν-sweep (bandwidth continuum, done):** tuning Burgers viscosity turned two regularity points into a curve; it *refuted* bandwidth as a *sufficient statistic* for advantage (V-shaped, not monotone).
- **Qubits** (3–6): more bandwidth + Hilbert space (changes params; needs re-matching + barren-plateau watch).
- **Measurement:** single / multi / ZZ, and the not-yet-implemented *probability vector* readout that the challenge explicitly requests.
- **input_scale** and **loss weighting:** secondary bandwidth / optimisation levers (the latter drove the earlier full-quantum collapse).
- **Encoding scheme:** the coordinate→qubit map (round-robin vs split vs all-on-all).

6.6 Tier 4 — Real CFD and mathematical depth

- **Lid-driven cavity (2-D incompressible Navier–Stokes):** the real-world CFD case the judges reward; the PDE-agnostic bench already supports multi-output, multi-equation systems. Literature suggests quantum may *lose* here — a valid "what is lost" finding.

Tier	Content	Est. compute	Promotes
2a	convergence (iterations)	$\sim$5 h	undertraining $\rightarrow$ converged
2b	3–5 seeds on headlines	$\sim$10 h	hypotheses $\rightarrow$ error-barred
2c	classical-front-end control	$\sim$0	removes diversity confound
3	ν-sweep + remaining knobs	$\sim$15 h	mechanism $\rightarrow$ sufficiency
4	real CFD + coefficient/BP analysis	TBD	breadth + math rigour

Table 6: Experimental roadmap with rough compute cost (single laptop, `default.qubit`). `lightning.qubit` may accelerate the simulation but currently breaks the higher-order autograd needed for u_{xx}.

- **Fourier coefficient reachability (Mhiri):** measure realised *coefficients* vs the reference, not just the frequency support, to test whether the depth-5 bandwidth shortfall (max freq 4, not 5) is the coefficient constraint biting.
- **Barren-plateau / trainability study:** systematic gradient-variance vs qubits/depth over many random initialisations; relation to the "black-hole" collapse.

6.7 The deliverable — from "design methodology" to a falsification protocol

The challenge asks (Task 4) for a problem-specific circuit-design methodology. Our data do not support one: with no robust quantum-specific advantage, a proxy$\rightarrow$bandwidth$\rightarrow$circuit prescription would be premature. What our study *does* yield — and what we offer as the transferable deliverable — is a **falsification protocol for claimed QAPINN advantages**, distilled from the three overclaims it killed:

1. **Train to convergence first** (L-BFGS, not Adam-only): most apparent advantages here were undertraining artifacts.
2. **Compare against the right control:** a param-matched *and* structurally-matched (same post-net) *and* depth-matched classical net — not an arbitrary MLP. Each stricter control killed a different apparent effect.
3. **Require multi-seed significance** ($n \geq 6$): an $n=3$ effect at $p=0.02$ here reversed at $n=6$.
4. **Separate features from model:** verify claims about the VQC's Fourier band on the VQC output, not the post-net-filtered model output.

Applied to our own results, this protocol converts three would-be positives into an honest null — which is precisely its value. A genuine design methodology becomes warranted only once an advantage survives all four gates on some problem; the future tiers below are how one would look for that.

7 What Is Currently Left Out (Scope Boundaries)

For transparency, the present study explicitly does *not* include (note: the statistics/convergence/knob items of the earlier draft are now *done* — see the converged and larger-model sections — and are removed from this list):

- **Breadth of PDEs:** only Burgers and heat; no real CFD (lid-driven cavity, though `cavity.py` is built), and the ν-continuum is single-seed.
- **Larger scales / more seeds:** the decisive comparisons are $n \leq 6$ with high seed variance; still-larger models and 5–8 seeds would firm the nulls.

- **Probability-vector readout:** the $\langle Z\rangle/\langle ZZ\rangle$ measurements are ablated, but the probability-vector observable the challenge lists is not implemented (it breaks the higher-order autograd needed for u_{xx}).
- **Coefficient-level Fourier analysis:** we measure realised frequency *support*, not the realised *coefficients* vs their reachable set (Mhiri).
- **Hardware / noise:** statevector simulation only; no shot noise, no device runs; "parameter reduction" is not "resource reduction" ($\sim$85$\times$ slower in simulation).
- **Validated design methodology:** because we find no robust quantum-specific advantage, a proxy→bandwidth→circuit prescription is not warranted by our data; what we offer instead (below) is a *falsification protocol* for claimed advantages.

8 Conclusion

We have a verified bench, a five-dimension XAI suite, and — crucially — an adversarial self-audit that repeatedly overturned our own readings. Across two model scales, three control types (matched, structural, depth-matched), and multi-seed statistics, the QAPINN offers *no statistically significant* advantage over a fairly-chosen classical baseline: at small scale classical wins, at resolved scale a depth-matched classical net ties the QAPINN, and no robust entanglement effect survives seed expansion. The "VQC = Fourier band = depth" fact is verified but governs the quantum features, not the full model. The honest contribution is a rigorous negative and the methodology behind it — three apparent advantages each dissolved under the appropriate fair control, and each would have been a false positive without it. The open work is to test still-larger scale, more seeds, and other PDEs before asking whether any quantum-specific advantage — or design methodology — is warranted.

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